



InvoSight

QR-based POS & Invoice Analyzer

Software Project Management

29 September 2026

Our Lecturer: Dr. Carl JR Perez

Prepared by:

Top Keitaro (Leader)
Chui Vanenteakbot
Sorn Kunwath
Lao David
Keo Ratanak

TABLE OF CONTENTS

1.	Project Overview.....	1
2.	Background / Introduction.....	1
3.	Goals and Objectives.....	1
4.	Our Service	2
5.	Project Scope.....	3
6.	Software Requirements	4
6.1.	Functional Requirements.....	4
6.2.	Non Functional Requirements.....	5
7.	Technology Stack	7
8.	System Architecture / Design	8
9.	Entity Relational Diagram	10
10.	Features and Screenshots.....	12
	Customer – Menu&Order.....	12
	Customer – Shopping Cart	13
	Customer – Item Customization.....	14
	Customer – Order Confirmation	15
	Cashier – Account Selection	16
	Cashier – PIN Authentication	17
	Cashier – New Order	18
	Cashier – Order & Progress.....	19
	Cashier – Receipt.....	20
	Admin – Dashboard Overview.....	21
	Admin – Menu Management (CRUD)	22
	Admin – Order Management (CRUD).....	23
	Admin – Create New Order	24
	Admin – Income & Expense Management.....	25
	Admin – Export Invoice	26
	Admin – Staff Account Management.....	27
11.	Work Plan / Timeline.....	28

12.	Budget and Resources	28
13.	Risk Management	30
14.	Expected Output / Deliverable	33
15.	Future Enhancement	34
16.	Meet our Team.....	36

1. Project Overview

InvoSight is a web based Point of Sale (POS) and QR ordering system designed for small and medium sized cafés and restaurants. It allows customers to scan table QR code using their smartphones, browse the digital menu, customize items and place orders without downloading a mobile application.

The system also helps cashiers manage incoming and walk in orders, update preparation statuses and confirm cash or KHQR payments. Administrators can manage menu items, categories, modifiers, cashier accounts and café tables while monitoring sales, expenses and business performance through dashboards and reports. Order and payment information is securely recorded in the database, allowing administrators to generate and download PDF receipts and sales reports when needed.

By combining customer ordering, cashier operations and administrative management into one platform, InvoSight aims to reduce manual work, minimize ordering errors, improve service efficiency and support better business decision making.

2. Background / Introduction

Many cafés and small restaurants still rely on paper based ordering and manual record keeping. These traditional methods can lead to incorrect orders, slower service and difficulties in tracking sales and daily business activities. As smartphone usage continues to grow, QR code technology provides an accessible and cost effective way to improve the ordering process without requiring customers to install a mobile application.

To address these challenges, InvoSight was developed as a web based system that combines QR ordering, Point of Sale (POS) operations, payment recording, receipt generation and sales analytics in one platform. Customers can browse the menu and place orders using their smartphones, while cashiers can manage incoming orders, preparation statuses and payments. Business owners can monitor sales, expenses and product performance through the administrator dashboard and reports.

By connecting these activities in one system, InvoSight helps reduce manual work, minimize ordering errors, improve service efficiency and support better business decision making.

3. Goals and Objectives

The goal of InvoSight is to develop a functional web based Point of Sale (POS) and QR ordering system that helps small and medium sized cafés and eateries manage their daily operations more efficiently. The system aims to streamline the ordering process, reduce manual

errors, improve service speed, and maintain organised records of orders, payments, and expenses. By providing sales reports and business performance insights, InvoSight enables business owners to monitor their operations and make informed decisions.

The objectives of InvoSight are to:

- Develop a QR code based ordering system that allows customers to access a digital menu, customise menu items, and place orders using their smartphones without installing a mobile application.
- Provide near real time order updates to cashiers through automatic polling, allowing incoming orders to be identified and processed efficiently.
- Implement an integrated Point of Sale (POS) system for managing walk in and QR orders, order statuses, payments, discounts, balances, and refunds.
- Store order, payment, and expense records in a database for future reference and record keeping.
- Generate downloadable receipts when requested while maintaining the related transaction information in the system.
- Generate daily, weekly, monthly, and yearly sales reports to help business owners monitor financial and operational performance.
- Provide product performance analytics to identify best selling items and support data driven business decisions.
- Provide administrative tools for managing menu items, categories, modifiers, cashier accounts, café tables, QR codes, settings, and activity records.
- Improve operational efficiency by reducing manual processes, minimizing ordering errors, and saving time for customers, staff, and business owners.

4. Our Service

InvoSight provides a web based ordering and café management service for customers, cashiers, and administrators. Customers can scan table QR code to browse the digital menu, customize items, place orders, and check their order status without installing an application. Cashiers can receive near real time order updates, create walk in orders, manage preparation statuses, record cash or KHQR payments, and generate receipts. Administrators can manage menu items, categories, modifiers, cashier accounts, café tables, QR codes, expenses, settings, and activity records. The system stores order, payment, and expense data in a database and provides sales reports and product performance analytics to support business monitoring and decision making.

5. Project Scope

In Scope	Out of Scope
Web based Point of Sale (POS) and QR ordering system.	Native Android or iOS mobile applications.
Table specific QR codes that allow customers to access the digital menu without installing an application.	Online food delivery and delivery route management.
Customer menu browsing, item customization, order placement, cancellation of pending orders, and order status tracking.	Integration with third party delivery platforms such as Food Panda or Grab Food.
Near real time order updates through automatic polling.	Instant updates through WebSockets or a dedicated real time notification service.
Cashier creation and management of walk in and QR orders.	Customer account registration and login.
Order workflow management, including Pending, Preparing, Ready, and Completed statuses.	Customer loyalty, membership, or reward programmers.
Cash and static KHQR payment recording with manual payment confirmation by authorized staff.	Automated online payment gateways or direct Bakong KHQR API integration.
Payment balance, refund calculations, and authorized payment voiding.	Automated bank payment verification and settlement.
Database storage of order, payment, expense, menu, table, and user records.	Permanent invoice numbering, invoice file storage, and a dedicated invoice archive.
Generation and download of receipts when requested.	Full accounting, tax filing, or payroll management.
Administrative management of menu items, categories, modifiers, cashier accounts, café tables, QR codes, settings, and activity records.	Inventory, stock, purchasing, and supplier management.
Expense recording and income versus expense monitoring.	Multi branch or franchise management.
Daily, weekly, monthly, and yearly sales reports with PDF and spreadsheet exports.	AI based sales prediction or demand forecasting.

Sales dashboards and product performance analytics, including best selling products.	Marketing automation, promotions, coupons, or email campaigns.
Separate administrator and cashier authentication with role based access control.	Advanced cybersecurity services such as penetration testing and automated fraud detection.
Operation through a web browser on supported desktop and mobile devices.	Guaranteed offline operation without access to the application server or network.

6. Software Requirements

6.1. Functional Requirements

Functional requirements define the actions that InvoSight must perform for each user role. The requirements are organized into Customer, Cashier, and Administrator functions.

- **Customer**

- Scan a table specific QR code to access the correct digital menu.
- Browse menu items and filter them by category.
- View item details, prices, availability, and images.
- Customise menu items using available modifiers and options.
- Add items to a cart and review the order before submission.
- Place an order without creating an account or installing an application.
- Receive an order number and view the submitted order summary.
- Monitor order progress through the Pending, Preparing, Ready, and Completed stages.
- Cancel an order while its status is still Pending.
- View the café's static KHQR payment code and the order amount.
- Select "I've Paid" to submit a KHQR payment claim for staff verification.

- **Cashier**

- Log in using their employee ID and six digit PIN.
- Log out securely when their work session is complete.
- View incoming QR and walk in orders in a central order queue.
- Receive near real time order updates through automatic polling.
- View order details, including the order number, table number, items, quantities, modifiers, notes, and total amount.
- Create orders manually for walk in customers.
- Edit eligible walk in orders while they are still Pending.
- Move orders through the Pending, Preparing, Ready, and Completed stages.
- Cancel eligible orders and provide a cancellation reason.

- Print order stickers containing the information required for preparation.
 - Record payments using cash or static KHQR.
 - Review and manually confirm customer KHQR payment claims.
 - Record the amount received and calculate any remaining balance or refund.
 - Generate and print receipts when requested.
 - View active orders and relevant order history.
- **Admin(shop owner)**
 - Log in using an administrator account and recover a forgotten password.
 - View a dashboard containing sales, order, and business performance information.
 - Receive near real time dashboard and order updates through automatic polling.
 - View order lists, order history, and detailed order information.
 - Create, update, deactivate, reactivate, and delete eligible menu items.
 - Manage menu categories, modifiers, and modifier options.
 - Create and manage cashier accounts.
 - Manage café tables and generate table specific QR codes.
 - Deactivate or reactivate tables and prevent tables with order history from being deleted.
 - Record, update, filter, and delete expense records.
 - View of income, expenses, net profit, and product performance analytics.
 - Filter sales information daily, weekly, monthly, and yearly periods.
 - Export business reports in PDF and spreadsheet formats.
 - View cashier and system activity records.
 - Manage café information, application settings, and administrator profile details.
 - Perform authorized payment void operations while preserving the related activity record.

6.2. Non Functional Requirements

Non functional requirements define the quality, security, usability, and operational standards expected from InvoSight. They describe how effectively the system should perform rather than the individual functions it provides.

1. Performance

Common pages and actions should respond within three seconds under normal café or demonstration usage. Customer orders, cashier polling, and dashboard updates

should not cause noticeable delays. Formal load testing is still required before claiming support for a specific number of simultaneous users.

2. Security

Administrator passwords and cashier PINs must be securely hashed before being stored. Role based access control must prevent cashiers from accessing administrator functions. The system must apply CSRF protection, authentication checks, input validation, and rate limiting where appropriate. Important actions, including login, logout, order cancellation, payment voiding, and record modification, should be recorded in activity logs.

3. Usability

The customer menu must be understandable without written instructions. The cashier interface must provide clear order information and controls that can be used efficiently during busy periods. The administrator interface must organise management tools, reports, and settings clearly.

4. Reliability

Successfully submitted orders and confirmed payments must be stored in the database and remain available after a page refresh. If a network request fails before the system confirms success, the user must be informed and allowed to retry. Near real time polling should recover automatically after a temporary connection failure.

5. Scalability

Administrators must be able to add more tables, QR codes, menu items, categories, modifiers, and cashier accounts without changing the source code. Multi branch support is not included in the current version but may be considered in future development.

6. Responsiveness

The customer menu must work on supported smartphone browsers. Cashier and administrator interfaces must remain usable on desktop and tablet screen sizes without hiding essential controls or information.

7. Availability

The current version is designed for local and controlled demonstration use. It remains available while the application server and database are running. Production

uptime, automated backups, disaster recovery, and service level guarantees have not yet been implemented or verified.

8. Data Integrity

Orders, order items, modifiers, payments, expenses, and status changes must be stored accurately in the database. Order status and payment status must be recorded separately. Cancelled orders and voided payments must remain available for historical and auditing purposes instead of being silently deleted. Additional transactions and concurrency hardening is required before production deployment.

9. Compatibility

The customer interface should work on current versions of major browsers, including Google Chrome, Safari, Microsoft Edge, and Mozilla Firefox. Customers must not be required to install a mobile application.

10. Maintainability

The application must follow Laravel's Model View Controller structure and use clearly organised routes, controllers, models, middleware, views, migrations, and tests. Automated tests and version control must be used to reduce regression when features are updated.

11. Privacy

Customers place orders anonymously, so customer accounts and personal profiles are not required. Administrator passwords and cashier PINs must never be stored as readable text. Cashier identification documents and other sensitive uploads require protected storage and restricted access before the system is used in production.

12. Print Compatibility

The system must provide browser based print layouts for order stickers and receipts. Final paper size, spacing, and printer settings must be tested using the specific thermal printer selected by the café; compatibility with every thermal printer is not guaranteed.

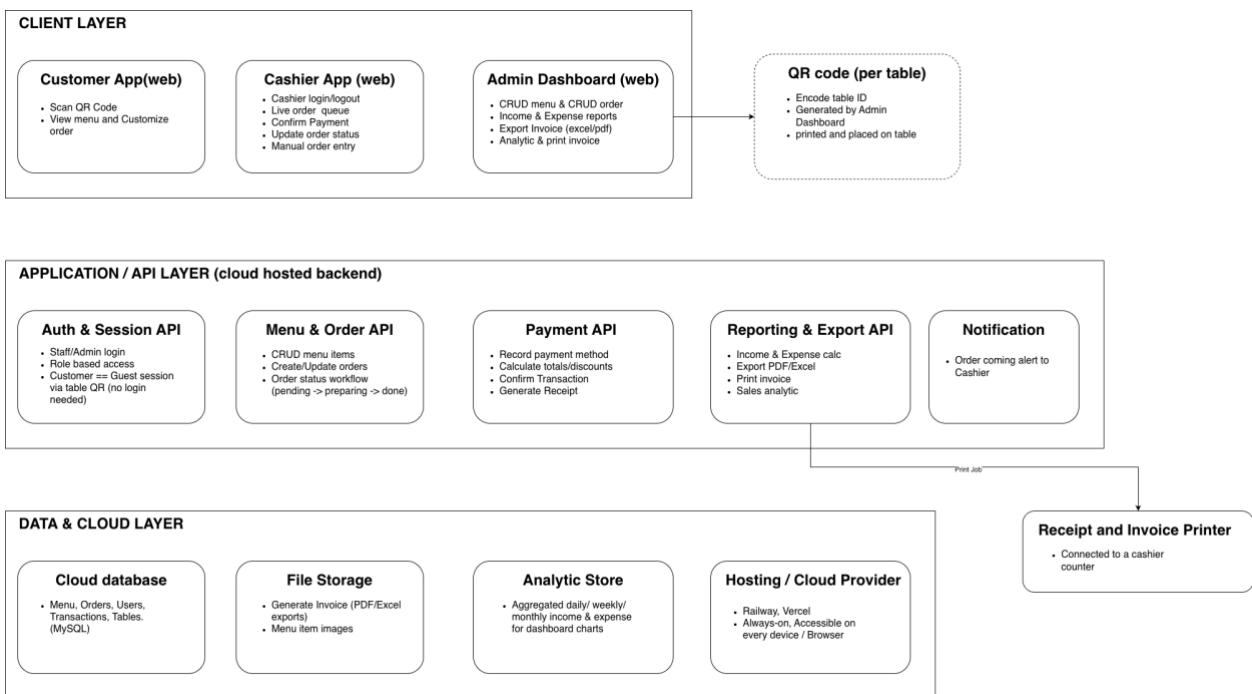
7. Technology Stack

The current InvoSight system uses the following technologies:

- Backend: PHP with Laravel 12

- Frontend: Laravel Blade, Tailwind CSS, and JavaScript
- Database: SQLite for development and testing, with MySQL support
- Authentication: Laravel Breeze for administrators and custom PIN authentication for cashiers
- Live Updates: AJAX polling using the Fetch API
- File Exports: DomPDF and PhPSpreadsheet
- QR Codes: Simple QrCode
- Testing and CI: PHPUnit and GitHub Actions
- Payment: Cash and static KHQR with manual confirmation
- Deployment: Local development server; production deployment has not yet been verified

8. System Architecture / Design



InvoSight uses a three tier structure within a single Laravel Model View Controller application. The three tiers are the Presentation Layer, Application Layer, and Data Layer. They are logically separated in the codebase but are not deployed as independent cloud services.

The Presentation Layer contains three browser based interfaces. Customers access the digital menu through a table specific QR code without creating an account. Cashiers use a separate interface to manage QR and walk in orders, update order statuses, confirm payments, and print receipts or order stickers. Administrators use the dashboard to manage menu items, categories, modifiers, cashiers, tables, QR codes, expenses, settings, reports, and activity records.

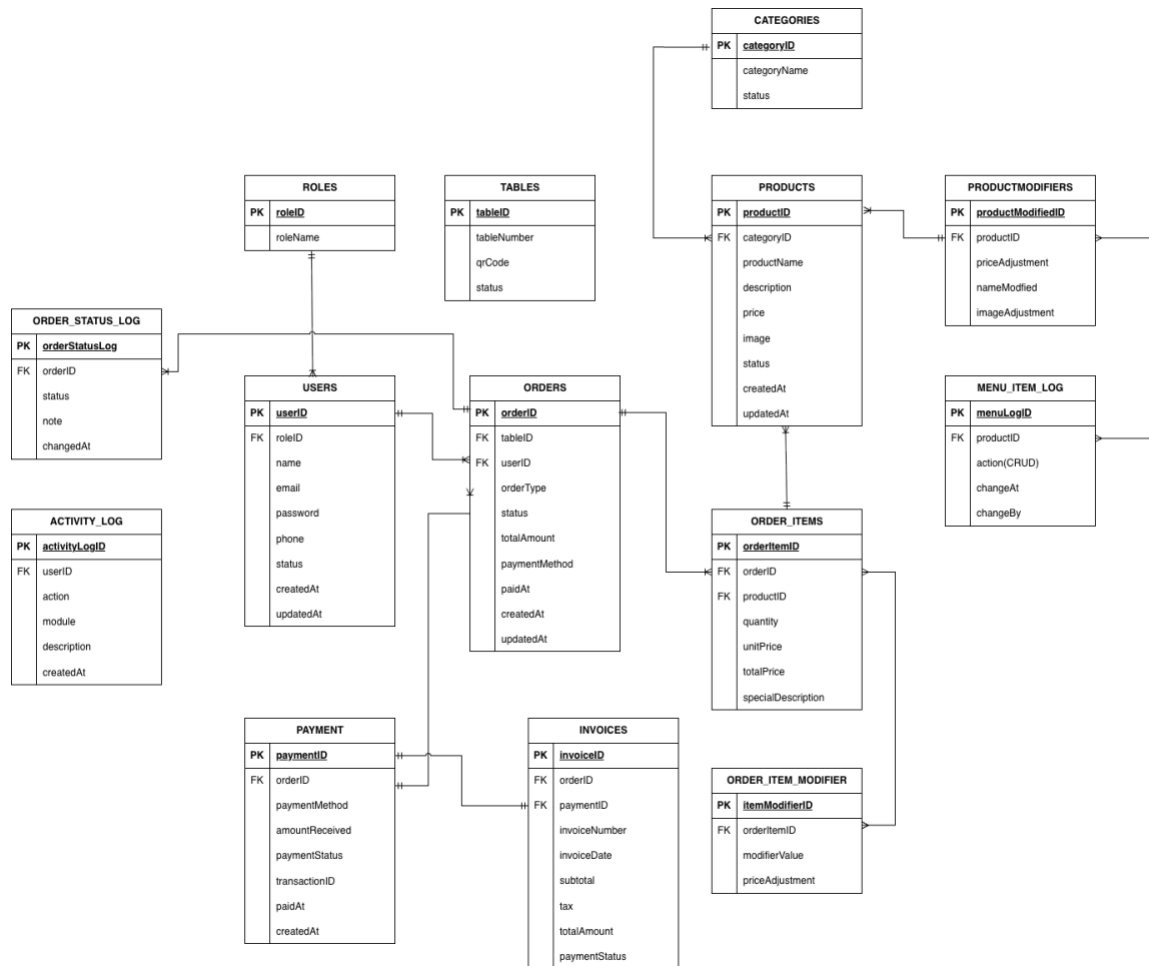
The Application Layer contains Laravel routes, controllers, middleware, models, and business rules. It handles administrator and cashier authentication, customer orders, payment recording, order status changes, expense management, reporting, and validation. Near real time updates are provided through JavaScript Fetch requests and AJAX polling rather than a separate notification service.

The Data Layer uses Laravel Eloquent to communicate with a relational database. SQLite is used in the current development and testing environment, while MySQL is supported through configuration. The database stores operational records such as users, menu items, tables, orders, payments, expenses, and activity logs. Reports and analytics are calculated from these records when requested; there is no separate analytics database.

Menu images and other uploaded files are stored through the application's file storage system. Receipts and reports are generated on demand using DomPDF and PhpSpreadsheet rather than stored as permanent invoice files. Printing is handled through browser based print layouts, not through a separate printer service.

All browser requests pass through the Laravel application before data is accessed or modified. The current system has been verified in a local development environment, while production cloud hosting has not yet been configured or verified.

9. Entity Relational Diagram



The Entity Relationship Diagram (ERD) illustrates the database structure of InvoSight, a QR based Point of Sale (POS) and ordering system. It defines the main entities, their attributes, and the relationships between them to support the system's core functions, including QR ordering, order management, payment processing, invoice generation, and administrative reporting. The database follows a relational model to ensure data consistency, minimize redundancy, and maintain referential integrity.

The Users entity stores all authenticated system users, including administrators and cashiers. Each user is assigned a role through the Roles entity, enabling role based access control within the system. The assigned user is also linked to each order to identify which staff member handled or managed the transaction. Additionally, the Activity Log entity records user activities, providing an audit trail for system operations.

The Tables entity represents the physical tables inside the café or restaurant. Each table is associated with a unique QR code that customers scan to access the digital menu. Every order

placed through QR ordering is linked to a specific table, allowing the system to identify the origin of each order without requiring customer registration.

The Categories entity organizes menu items into groups such as beverages, food, and desserts. Each category can contain multiple Products, while every product belongs to a single category. The Products entity stores information including product name, description, price, image, and availability status.

To support product customization, the ProductModifiers entity stores the available options for each product, such as sugar level, ice level, size, or additional toppings. When customers customize an item during ordering, the selected options are stored in the Order_Item_Modifier entity. This ensures that each customized order item is accurately recorded together with any additional price adjustments.

Customer purchases are managed through the Orders and Order_Items entities. An order represents a complete customer transaction associated with a table and handled by a staff member, while each order may contain multiple order items. The Order_Items entity records the purchased products, quantity, unit price, total price, and any special instructions entered by the customer.

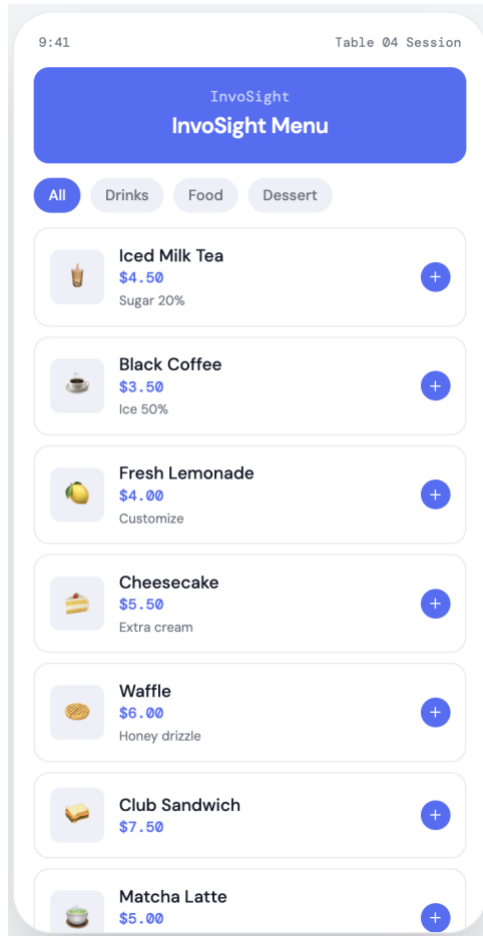
The Payment entity stores payment related information, including payment method, payment status, transaction ID, payment timestamp, and amount received. After a payment is successfully completed, the system generates an Invoice, which stores invoice details such as invoice number, invoice date, subtotal, tax, total amount, and payment status. This provides both customers and business owners with a permanent transaction record.

To improve accountability and maintain historical records, the system includes Order_Status_Log and Menu_Item_Log entities. The Order_Status_Log records changes to an order's status throughout its lifecycle, while the Menu_Item_Log records CRUD (Create, Read, Update, Delete) operations performed on menu items. These log tables support auditing, troubleshooting, and future analysis of system activities.

Overall, the ERD provides a structured representation of the InvoSight database by connecting users, menu management, ordering, customization, payments, invoices, and activity records into a single relational database. This design supports efficient daily operations while remaining scalable for future enhancements, such as customer accounts, loyalty programs, or multi branch management.

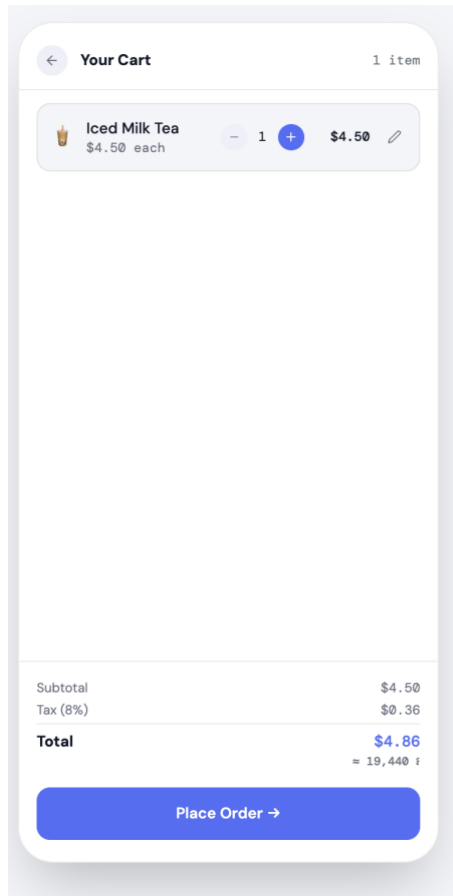
10. Features and Screenshots

Customer – Menu&Order



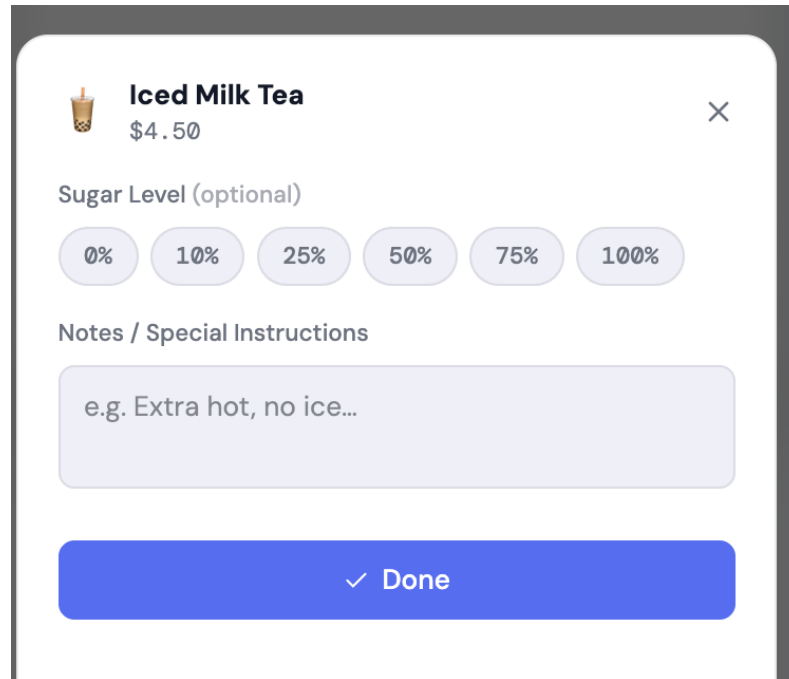
After scanning the table's QR code, the customer lands directly on the menu. No app download or account needed. Items are grouped by category (Drinks, Food, Dessert), and each item shows a quick customization tag (e.g. "Sugar 25%") before being added to the cart. A persistent cart bar at the bottom keeps the running total visible at all times, reducing the steps between browsing and paying.

Customer – Shopping Cart



The Shopping Cart allows customers to review their selected items before placing an order. Each item displays its quantity, unit price, and subtotal, while the plus and minus buttons enable customers to quickly adjust quantities without returning to the menu. The edit button allows customers to modify customization options such as sugar level or special instructions. At the bottom of the page, the system automatically calculates the subtotal, tax, and final payable amount in both US Dollars (USD) and an estimated Cambodian Riel (KHR) equivalent, giving customers a clear understanding of their payment before confirming the order.

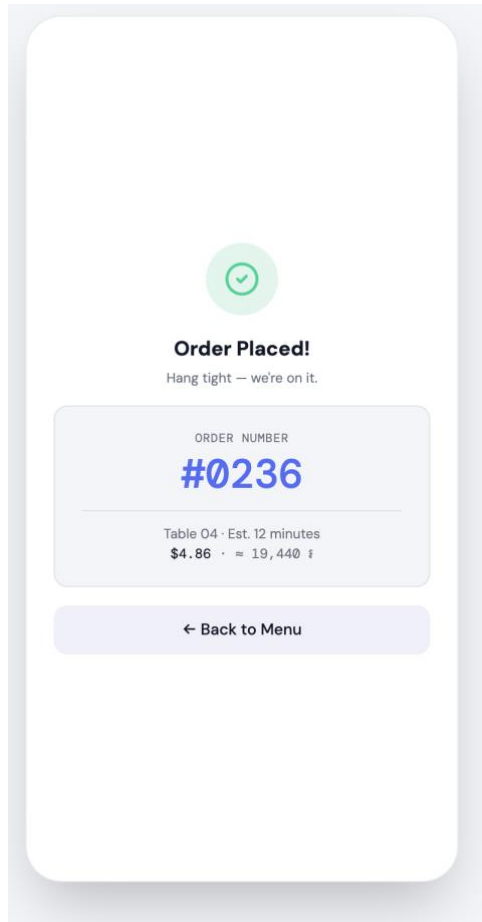
Customer – Item Customization



The screenshot shows a mobile app interface for customizing an item. At the top, there's a header for "Iced Milk Tea" with a price of "\$4.50" and a close button (X). Below this, the "Sugar Level (optional)" section features six buttons: 0%, 10%, 25%, 50%, 75%, and 100%. The 100% button is highlighted. Underneath is a text input field labeled "Notes / Special Instructions" with the placeholder text "e.g. Extra hot, no ice...". At the bottom is a large blue button with a checkmark and the text "Done".

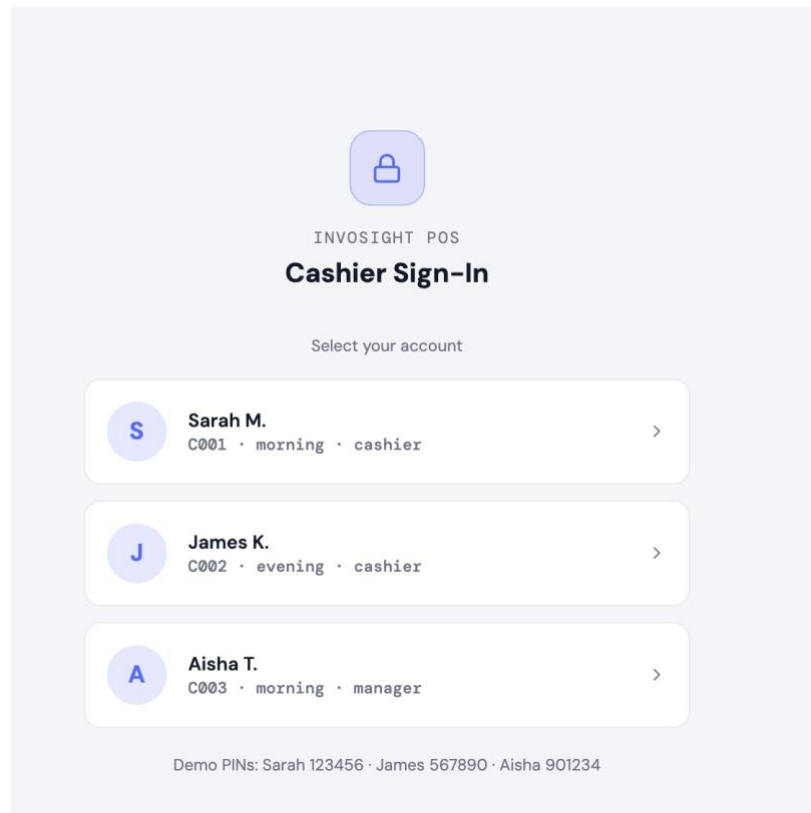
The Item Customization window allows customers to personalize menu items before adding them to their order. Customers can optionally select their preferred sugar level, ranging from 0% to 100%, and provide additional notes or special instructions, such as requesting less ice or extra hot beverages. These customization details are attached directly to the selected order item, ensuring that staff receive accurate preparation instructions while allowing customers to tailor their orders according to their preferences.

Customer – Order Confirmation



After an order has been successfully submitted, the Order Confirmation page provides immediate feedback to the customer. The page displays a unique order number, table number, estimated preparation time, and the total amount in both USD and estimated KHR. This confirmation reassures customers that their order has been received successfully while providing a reference number that can be used by both customers and staff during order preparation. The "Back to Menu" button also allows customers to continue browsing and place additional orders if desired.

Cashier – Account Selection



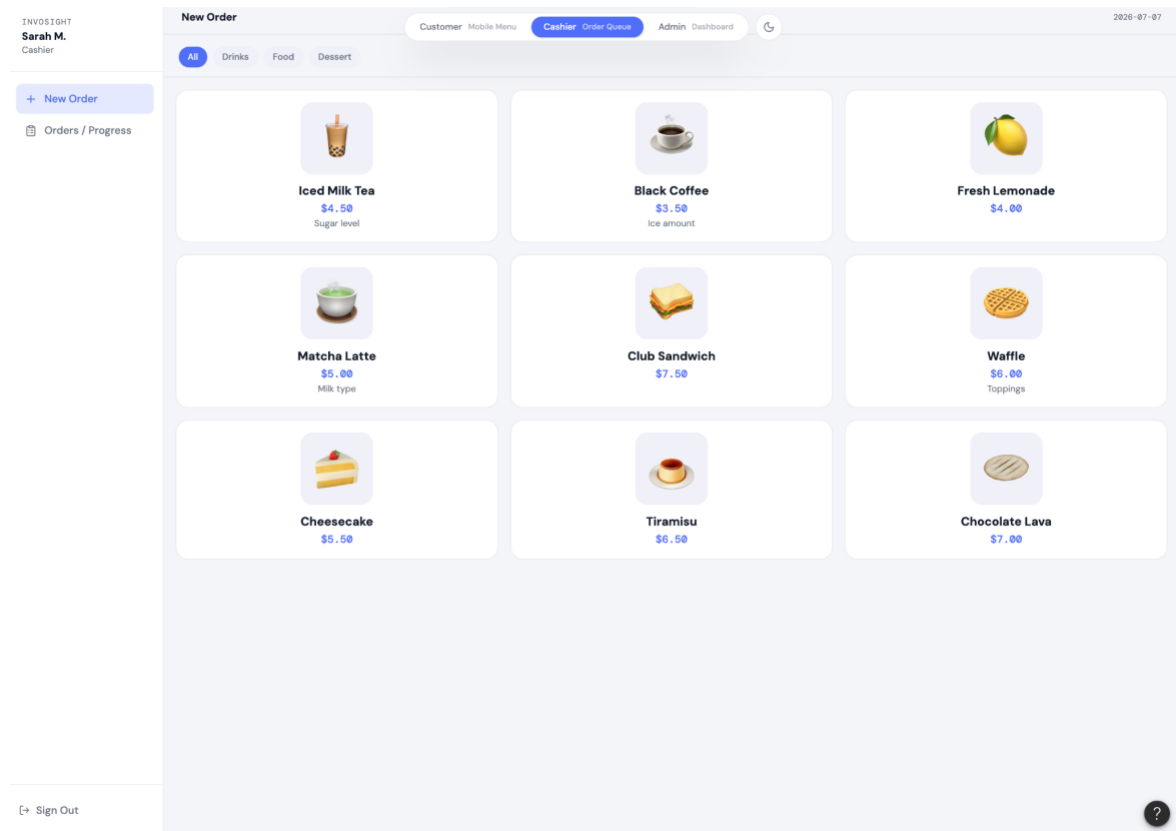
The Account Selection screen allows staff members to choose their registered account before accessing the Point of Sale (POS) system. Each account displays the staff member's name, employee ID, work shift, and assigned role, enabling quick identification before authentication. This approach supports role based access control by ensuring that only authorized personnel can access the system while allowing multiple cashiers or managers to use the same POS terminal.

Cashier – PIN Authentication

The screenshot displays the 'Cashier Sign-In' interface for INVOSIGHT POS. At the top, there is a lock icon and the text 'INVOSIGHT POS' and 'Cashier Sign-In'. Below this, a user profile for 'Sarah M.' (ID: C001, role: cashier) is shown with a 'Change' link. The main section prompts the user to 'Enter your 6-digit PIN' with a visual indicator of six dots (five filled, one empty). A 'Show PIN' toggle is present. Below the prompt is a numeric keypad with buttons for digits 1-9, 0, and a backspace icon. At the bottom, demo PINs are listed: Sarah 123456, James 567890, and Aisha 901234.

After selecting an account, the cashier must enter a six digit Personal Identification Number (PIN) to access the system. The numeric keypad provides a simple and efficient authentication method suitable for touchscreen POS terminals, while the optional "Show PIN" feature allows users to verify their input. This authentication process improves security by preventing unauthorized access while maintaining a fast login experience for staff during busy business hours.

Cashier – New Order



The New Order screen enables cashiers to manually create customer orders for walk in customers who do not use QR ordering. Cashiers can browse menu categories, select products, adjust quantities, specify table numbers, and enter special instructions or customization options such as sugar level. The system automatically calculates the total amount and displays the order summary before submission. This feature ensures that both manual orders and QR orders are processed through the same centralized ordering system, maintaining consistency and reducing order management complexity.

Cashier – Order & Progress

The screenshot displays the 'Orders & Progress' interface for a cashier named Sarah M. The interface is divided into two main sections: a left sidebar for order management and a right pane for detailed order information.

Left Sidebar (Orders & Progress):

- Active (3):**
 - #0231 (PENDING):** Table 04, 2x Iced Milk Tea, 1x Black Coffee, 09:14, \$12.50.
 - #0230 (PREPARING):** Table 02, 1x Fresh Lemonade, 1x Cheesecake, 09:08, \$9.50.
 - #0229 (READY):** Table 07, 3x Black Coffee, 13:01, \$10.50. This order is highlighted.
- COMPLETED (3):**
 - #0228 - Table 01, \$12.50
 - #0227 - Table 05, \$18.50
 - #0226 - Table 03, \$11.00

Right Pane (Order #0229):

- Table 07 · 13:01**
- Order #0229** (READY, 2:36)
- Auto-completing in 2:36. Use the button below to complete manually.**
- ORDER ITEMS:**
 - Black Coffee:** \$3.50 x 3 = \$10.50 (Extra hot)
- Subtotal:** \$10.50
- Tax (8%):** \$0.84
- Total:** \$11.34
- Buttons:** Collect & Pay, Print Receipt

The Orders & Progress screen allows cashiers to monitor and manage all customer orders throughout the ordering process. Orders are categorized by their current status, such as Pending, Preparing, Ready, and Completed, enabling staff to quickly identify which orders require attention. Selecting an order displays detailed information, including ordered items, special instructions, subtotal, tax, and total payment. Cashiers can also collect payments and print receipts directly from this interface, ensuring an efficient workflow from order preparation to customer checkout.

Cashier – Receipt

Receipt

InvoSight Café

123 Coffee Lane · 555-0100

Order #0229 · Table 07

2026-07-07 13:01

3× Black Coffee

↳ Note: Extra hot

Subtotal

Tax 8%

TOTAL

in KHR

\$10.50

\$0.84

\$11.34

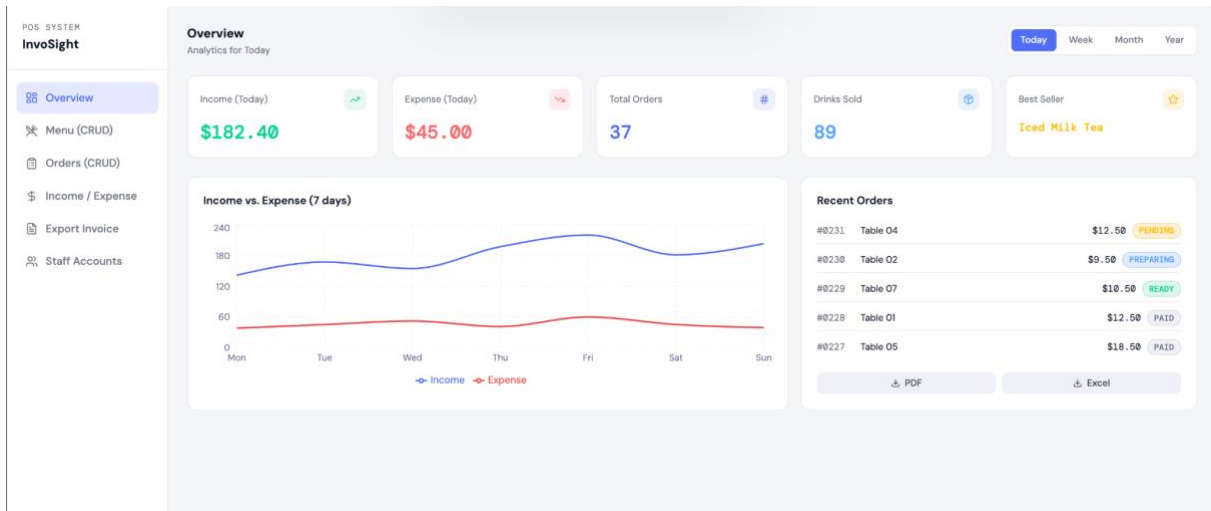
45,360 ₭

Thank you! Come back soon 🍷

Print (simulated)

The Receipt window displays a digital representation of the customer's receipt before printing. It includes essential transaction information such as the café name, order number, table number, purchased items, special instructions, subtotal, tax, total amount, and the equivalent value in Cambodian Riel (KHR). By allowing cashiers to review receipt details before printing, the system helps reduce printing errors and ensures customers receive accurate transaction records.

Admin – Dashboard Overview



The Dashboard Overview provides administrators with a summary of the business's daily performance through key performance indicators (KPIs) and graphical reports. It displays important information such as total income, expenses, number of orders, products sold, and the best selling menu item. The dashboard also includes an income versus expense chart and a list of recent orders, allowing administrators to quickly monitor business performance and operational status. Time filters such as Today, Week, Month, and Year enable users to analyze business trends over different reporting periods.

Admin – Menu Management (CRUD)

POS SYSTEM
InvoSight

Overview

Menu (CRUD)

Order Mgmt

Income/Expense

Export Invoice

Staff Accounts

Menu Management

10 items - 9 available

Customer Mobile Menu Cashier Order Queue Admin Dashboard

+ Add Item

Q Search...

ITEM	CATEGORY	PRICE	STATUS	
 Iced Milk Tea Sugar level	DRINKS	\$4.50	AVAILABLE	 
 Black Coffee Ice amount	DRINKS	\$3.50	AVAILABLE	 
 Fresh Lemonade	DRINKS	\$4.00	AVAILABLE	 
 Matcha Latte Milk type	DRINKS	\$5.00	AVAILABLE	 
 Club Sandwich	FOOD	\$7.50	AVAILABLE	 
 Waffle Toppings	FOOD	\$6.00	AVAILABLE	 
 Pasta Carbonara	FOOD	\$10.00	UNAVAILABLE	 
 Cheesecake	DESSERT	\$5.50	AVAILABLE	 
 Tiramisu	DESSERT	\$6.50	AVAILABLE	 
 Chocolate Lava	DESSERT	\$7.00	AVAILABLE	 

The Menu Management module allows administrators to manage all menu items available in the system. Administrators can add new products, edit existing menu information, update prices, assign product categories, and change product availability status. A search function is provided to quickly locate specific menu items, while editing and deleting actions simplifying menu maintenance. This feature ensures that the digital menu remains accurate and up to date for both customers and cashiers.

Admin – Order Management (CRUD)

POS SYSTEM InvoSight

Order Management
6 total - 3 active

Customer Mobile Menu Cashier Order Queue **Admin Dashboard**

+ Add Order

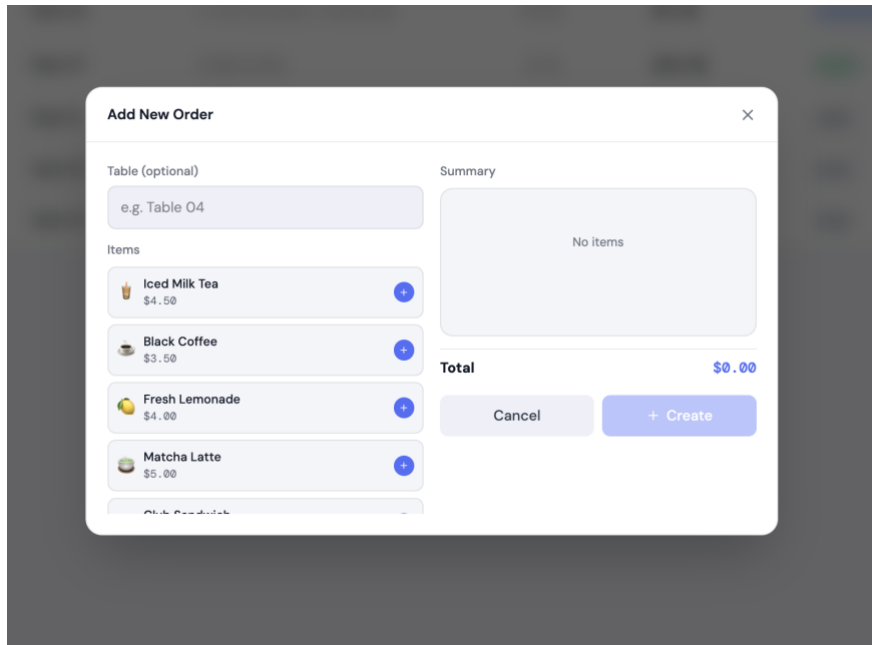
Q Search...

All Pending Preparing Ready Paid

ORDER	TABLE	ITEMS	TIME	TOTAL	STATUS	ACTIONS
#0231	Table 04	2x Iced Milk Tea, 1x Black Coffee	09:14	\$12.50	PENDING	Edit Delete
#0230	Table 02	1x Fresh Lemonade, 1x Cheesecake	09:08	\$9.50	PREPARING	Edit Delete
#0229	Table 07	3x Black Coffee	13:01	\$10.50	READY	Edit Delete
#0228	Table 01	1x Club Sandwich, 1x Matcha Latte	14:52	\$12.50	PAID	Edit Delete
#0227	Table 05	2x Waffle, 1x Tiramisu	18:39	\$18.50	PAID	Edit Delete
#0226	Table 03	2x Cheesecake	19:22	\$11.00	PAID	Edit Delete

The Order Management module enables administrators to monitor and manage all customer orders from a centralized interface. Orders can be searched, filtered by their status, and viewed with details such as table number, ordered items, order time, total amount, and payment status. Administrators can also edit, update, or remove orders when necessary, ensuring accurate order tracking and efficient management of customer transactions.

Admin – Create New Order



The screenshot shows a modal window titled "Add New Order" with a close button (X) in the top right corner. The window is divided into two main sections: "Table (optional)" and "Items".

Table (optional): A text input field containing "e.g. Table 04".

Items: A list of menu items, each with a small icon, the item name, the price, and a blue "+" button to add it to the order:

- Iced Milk Tea \$4.50
- Black Coffee \$3.50
- Fresh Lemonade \$4.00
- Matcha Latte \$5.00

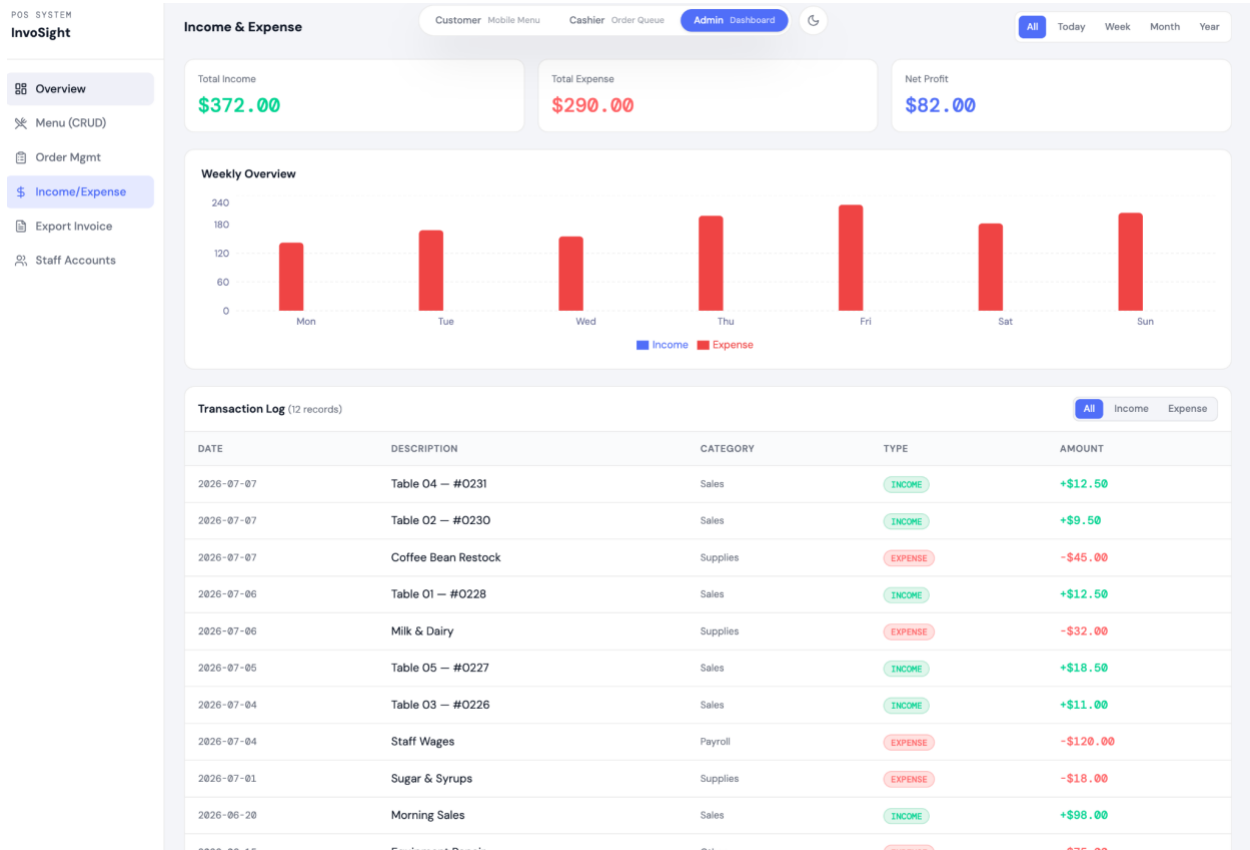
Summary: A large light blue box containing the text "No items".

Total: A label "Total" followed by the calculated amount "\$0.00".

Buttons: At the bottom right, there are two buttons: a light blue "Cancel" button and a blue "+ Create" button.

The Create New Order window allows administrators to manually create customer orders directly from the dashboard. Administrators can assign an optional table number, select menu items, review the order summary, and verify the calculated total amount before confirming the order. This feature provides flexibility for handling special cases or manually entering customer orders while maintaining consistency with the system's ordering process.

Admin – Income & Expense Management



The Income & Expense module provides administrators with a comprehensive financial overview of the business. The system summarizes total income, total expenses, and net profit while presenting graphical reports to visualize financial performance over selected periods. A transaction log records every income and expense entry, including sales revenue, operational expenses, and payroll costs. This module assists business owners in monitoring cash flow and evaluating the overall financial health of the business.

Admin – Export Invoice

POS SYSTEM
InvoSight

Customer Mobile Menu Cashier Order Queue **Admin Dashboard**

Export Invoice

Session Filter: **All** Morning Afternoon Evening

Select Order: **#0231 — Table 04 (09:14) \$12.50**

Order	#0231
Table	Table 04
Time	09:14
Status	PENDING
Total (incl. tax)	\$13.50 = 54,000 ₭

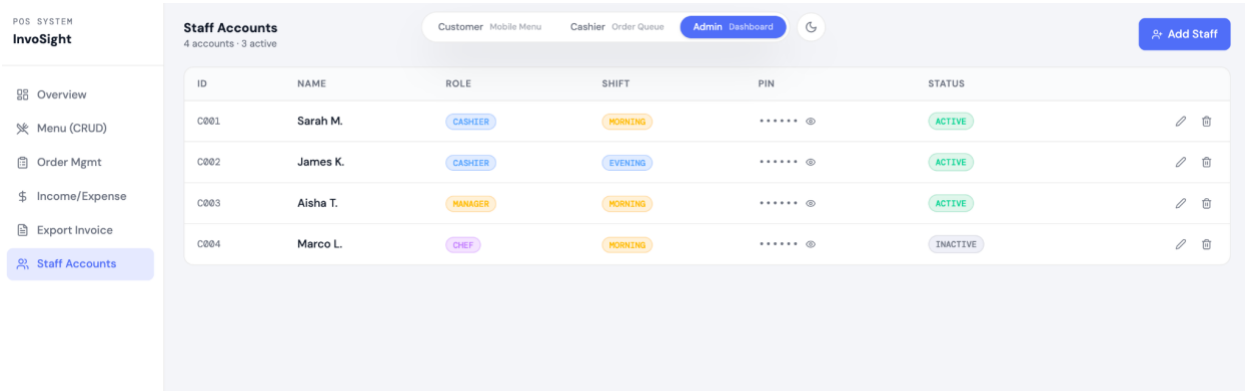
Export PDF **CSV**

InvoSight Café
123 Coffee Lane • 555-0100
TAX INVOICE

Invoice #	INV-0231
Date	2026-07-07
Session	All
Table	Table 04
2× Iced Milk Tea	\$9.00
↳ Sugar: 50%	
1× Black Coffee	\$3.50
Subtotal	\$12.50
GST 8%	\$1.00
TOTAL	\$13.50
in KHR	54,000 ₭

The Export Invoice module allows administrators to retrieve and export completed transaction records. Users can filter invoices by work session, select a specific order, preview the invoice details, and export the document in PDF or CSV format. The exported invoice contains essential transaction information, including order number, table number, purchased items, tax, total amount, and invoice date. This feature supports financial reporting, record keeping, and business documentation.

Admin – Staff Account Management



POS SYSTEM
InvoSight

Staff Accounts
4 accounts · 3 active

Customer Mobile Menu Cashier Order Queue Admin Dashboard

Add Staff

ID	NAME	ROLE	SHIFT	PIN	STATUS	
C001	Sarah M.	CASHIER	MORNING	***** @	ACTIVE	 
C002	James K.	CASHIER	EVENING	***** @	ACTIVE	 
C003	Aisha T.	MANAGER	MORNING	***** @	ACTIVE	 
C004	Marco L.	CHEF	MORNING	***** @	INACTIVE	 

The Staff Account Management module enables administrators to manage employee accounts within the POS system. Administrators can create, edit, or deactivate staff accounts while assigning roles, work shifts, PIN authentication, and account status. The module displays important employee information such as staff ID, role, shift schedule, and account availability, allowing administrators to efficiently control system access and maintain secure role based authentication for all employees.

11. Work Plan / Timeline

PHASE	TASK TITLE	TASK OWNER	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
1	PLANNING																	
1.1	Project Overview	Rathanak																
1.2	Background / Introduction	Rathanak																
1.3	Goals and Objectives	David																
1.4	Our Service	David																
1.5	Project Scope	David																
1.6	Software Requirements	Vanenteakbot																
1.7	Technology Stack	Vanenteakbot																
1.8	System Architecture / Design	Keitaro																
1.9	Features and Screenshots	Keitaro																
1.10	Work Plan / Timeline	Kunwath																
1.11	Budget and Resources	Kunwath																
1.12	Risk Management	Kunwath																
1.13	Expected Output / Deliverable	Kunwath																
2	DESIGN																	
2.1	User Flow Diagram	Keitaro																
2.1	Wireframe Design	Kunwath																
2.1	UX/UI Design	David																
2.1	Database Design	Vanenteakbot																
2.1	System Architecture / Design	Keitaro																
3	DEVELOPMENT																	
3.1	Frontend Development	Rathanak																
3.2	Backend Development	Vanenteakbot																
3.3	Database Development	Vanenteakbot																
3.4	QR ordering Module	Kunwath																
3.5	POS Module	David																
3.6	Dashboard Module	Keitaro																
4	TESTING																	
4.1	Functional Testing	Everyone																
4.2	User Acceptance Testing	Everyone																
4.3	Bug Fixing	Everyone																
5	DEPLOYMENT																	
5.1	Documentation	Everyone																
5.1	System Deployment	Vanenteakbot																
5.1	Final Presentation	Everyone																

12. Budget and Resources

To build InvoSight, we have estimated the costs required. It covers hosting, development tools, and hardware needed throughout the project. Most tools are free or open source, keeping overall costs low. A small hosting fee and testing hardware make up the bulk of the budget.

Hosting			
Item	Description	Type	Cost
Railway app server	Hosts the Laravel application and serves it online, upgraded plan for reliable uptime and automated database backups.	Paid	\$ 20 /month
Domain name	Provides a custom URL for clients to	Paid	\$ 10

	access the system.		/year
SSL certificate	Encrypt data between users and the server.	Free	\$ 0

Development Tools			
Item	Description	Type	Cost
Laravel framework	Main PHP framework used to build the entire system.	Free/OSS	\$ 0
MySQL database	Store all data including menus, orders, and users.	Free/OSS	\$ 0
Tailwind CSS	Utility first CSS framework for styling the fronted UI.	Free/OSS	\$ 0
Laravel Breeze	Handles authentication (login/logout) for staff and admin.	Free/OSS	\$ 0
DomPDF	Generates printable PDF receipts from order data.	Free/OSS	\$ 0
Meatwebsite Excel	Exports sales to Excel format.	Free/OSS	\$ 0
PHP qr Code Library	Generates QR codes for tables.	Free/OSS	\$ 0

Hardware			
Item	Description	Type	Cost
Development laptops	Used by the team to write and test the system.	Existing	\$ 0
Test smartphones	Used to test QR scanning and mobile ordering UI	Existing	\$ 0
Thermal printer	Used for testing receipt printing.	Paid	\$ 80
POS machine	Used for testing cashier workflow and order management during development.	Paid	\$ 600

Contingency

Item	Description	Type	Cost
Contingency	Buffer for unexpected minor costs (replacements, testing consumables, printing, etc.).	Buffer	\$ 500

Personnel Cost			
Team Member	Position	Monthly Salary	4 Months
Top Keitaro	Project Manager & System Architect	\$ 900	\$ 3600
Chui Vanenteakbot	Software Developer & Documentation	\$ 700	\$ 2800
Sorn Kunwath	Software Developer & Documentation	\$ 700	\$ 2800
Lao David	Software Developer & Documentation	\$ 700	\$ 2800
Keo Ratanak	Software Developer & Documentation	\$ 700	\$ 2800

Hosting	\$ 90
Development Tools	\$ 0
Hardware	\$ 680
Contingency	\$ 500
Personnel Cost	\$ 14,800
Total Budget	\$16,100

13. Risk Management

- **Project management Risks**

Risk	Mitigation	Level
------	------------	-------

Scope creep adding features mid build.	Lock feature list before development begins, as a signed agreement. All new ideas go to a V2 wishlist. No new features during development without team consensus.	High
Team member becomes unavailable	Share project documentation through Google Docs and regularly upload completed work to GitHub. This allows other team members to continue the unfinished tasks with minimal disruption.	Medium
Project schedule delays	Review progress weekly and adjust the timeline when necessary	Medium

- **Technical Risks**

Risk	Mitigation	Level
AJAX polling causes server overload under many simultaneous orders.	Set the polling interval to 5–10 seconds (not less). Test with simulated concurrent orders. Upgrade interval if needed.	High
Databases get corrupted or data is lost.	Use managed MySQL on Railway which handles automated backups. Also export a manual backup at the end of each development week.	High
Railway free tier limits or goes down during demo or testing.	Keep a fully working localhost version as backup. Prepare a demo using localhost if the Railway is unavailable on demo day.	High
Unable to connect with KHQR payment API	Start API integration early, thoroughly test all payment functions using the provider's documentation or sandbox, and resolve integration issues before deployment.	High

- **Operational Risks**

Risk	Mitigation	Level
Employees are unfamiliar with the system	Provide user training before deployment and prepare a simple user guide explaining the main system functions. During the first few days of operation,	Medium

	provide support to help staff become familiar with the ordering, payment, and reporting processes.	
KHQR payment service is unavailable	Allow cashiers to verify payments manually using the customer's payment receipt or transaction reference, then synchronize the payment once the service is available again.	High
High customer traffic slows the system	Optimize system performance, monitor server resources, and upgrade hosting resources if customer traffic increases.	High
Receipt does not print correctly on the thermal printer	Test printer compatibility before deployment, provide digital receipt previews, and allow receipt reprinting if printing fails.	Medium
Cashier or administrator account credentials are compromised	Enforce strong passwords, role based access control, and secure password hashing to protect user accounts.	High
Cancelled orders are accidentally deleted from the database	Use soft delete functionality and maintain audit logs so deleted records can be recovered if necessary.	Medium
PDF or Excel reports generate incorrect information	Validate exported data against the database and test report generation before deployment.	Low

Although we identified several potential risks during the project, these are the five most critical risks because they have the greatest impact on project development and business operations.

Rank	Risk	Risk Level
1	Unable to connect with KHQR payment API	High
2	Databases get corrupted or data is lost.	High
3	High customer traffic slows the system	High
4	Scope creep adding features mid build.	High
5	KHQR payment service is unavailable	High

14. Expected Output / Deliverable

No	Deliverable	Description	Type
1	Customer QR Ordering Interface	Mobile friendly web page that opens on QR scan. Shows menu, customization options, order review, No app download required.	Web Module
2	Cashier Dashboard	Real time order queue with table numbers, sorted by time. Buttons to start preparing, confirm payment (cash/KHQR), add/edit/cancel orders. Print receipt.	Web Module
3	Admin Panel	Full CRUD for menu items (with image, modifiers, deactivate toggle). CRUD for orders. View real time sales dashboard. Manage cashier accounts.	Web Module
4	Sales Reports & Analytics	Filterable sales reports by day/week/month/year. Charts for top items and revenue. Cashier activity log with login/logout and actions.	Report
5	PDF Receipt	Printable receipt generated by DomPDF. Contains order ID, table number, items, modifiers, quantity, cashier name, date/time, and total.	Document
6	PDF Report Export	Admin can export any filtered sales report as a formatted PDF document for record keeping or sharing with business owners.	Document
7	Excel Report Export	Admin can export any filtered sales report as a .xlsx file for further analysis in spreadsheet applications.	Document
8	QR Codes for Tables	Printed and laminated QR codes for each café table. Each QR encodes the table number and links directly to the ordering page.	Asset

9	Database Schema & Diagram	Full ERD showing all tables: orders, order_items, menu_items, modifiers, users, cashier_logs, and relationships.	Document
10	System Wireframes	UI wireframes for all 3 dashboards (customer mobile, cashier, admin) showing screen flow and key interactions.	Document
11	User Manual	Step by step guides for cashiers (how to use the dashboard, confirm payments, print receipts) and admins (menu management, reports).	Document
12	Add the Service Agreement	A signed agreement between the development team and the client confirming that the client subscribes to system access at \$150/month. The client receives a fully functional hosted system but does not own or receive the source code at any time. The monthly fee covers hosting, maintenance, and basic support. Either party may terminate with 30 days written notice.	Agreement
13	Deploy live system	The actual running application is hosted on Railway with domain configured, SSL active, and all accounts (admin + cashier) set up and ready to use.	System

15. Future Enhancement

The current version of InvoSight fulfils the core requirements for a controlled school demonstration. However, further improvements could strengthen its security, reliability, scalability, customer experience, and business capabilities. The following enhancements are proposed for future development.

No.	Enhancement	Description	Priority
1	Production Security Hardening	Strengthen public order protection, input validation, payment transaction handling, database constraints, request rate limiting, and private file storage before deploying the system in a production environment.	High
2	Production Hosting and Backups	Deploy InvoSight to a suitable cloud server with a domain name and HTTPS. Configure automated database backups, system monitoring, and	High

		scheduled background tasks to improve availability and data recovery.	
3	Automated KHQR Payment Integration	Integrate a supported Bakong KHQR payment API, subject to API access and provider requirements, to generate payment-specific QR codes, verify transactions automatically, and update payment records after successful confirmation.	High
4	Inventory Management	Introduce ingredient and product stock tracking, automatic stock deduction after completed orders, low-stock alerts, and purchasing and supplier records to help cafés manage inventory more efficiently.	High
5	Enhanced Customer Order Tracking	Improve the existing customer order-status page with estimated preparation times, clearer order stages, more reliable status updates, and optional browser notifications when an order is ready.	Medium
6	Kitchen Display System	Develop a dedicated kitchen display for viewing incoming orders, preparation details, order priorities, and completion status to improve coordination between cashiers and kitchen staff.	Medium
7	Invoice Management	Extend the existing receipt and document functionality with permanent invoice numbers, stored invoice files, invoice history, and options to retrieve and reprint invoices when required.	Medium
8	Customer Accounts and Loyalty	Introduce optional customer accounts with saved order history, favourite items, loyalty points, discounts, and promotional rewards to encourage customer retention.	Medium
9	Multi-Branch Management	Extend the system to support multiple café branches, allowing owners to manage branch-specific staff, menus, expenses, sales, and reports through a centralised platform.	Low
10	Advanced Analytics	Extend existing reports with sales forecasting, peak-hour analysis, item profitability, expense trends, and detailed performance comparisons to support business planning and decision-making.	Low

These enhancements are proposed improvements rather than features currently included in the implemented system. They could be introduced gradually according to business needs, available resources, and implementation priorities. Security improvements and reliable deployment should be addressed before the system is used in a production environment, while the remaining enhancements could be developed in later versions of InvoSight.

16. Meet our Team

Team Members:



Top Keitaro (Project Leader, Database Designer & UI/UX Designer)
tp.keitaro.works@gmail.com

Quotes: "*Stop Complaining. Start Doing*"



Chui Vanenteakbot (Laravel Backend Developer, Features planner)

chuibot5@gmail.com

Quotes: "Messy counter, messy data. Organized system, thriving business."



Sorn Kunwath (Project Planning and Document)

sornkunwath123@gmail.com

Quotes: "No time, no money, no project."



Lao David (Documenting, idea and project back testing)

laodavid205@gmail.com

Quotes: "Give everything you got but not *GIVE UP.*"



Keo Ratanak (Frontend developer)

Keoratanak81@gmail.com

Quotes: "Simple interfaces create the best user experiences."